C12 Alkanes – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Crude Oil and Cracking **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Crude oil				Mixture of mostly alkanes
2	Method used to separate crude oil				Fractional distillation
3	Property fractions are separated by				Boiling point
4	Hottest part of the fractionating column				Bottom
5	Where the shortest chains are collected				Top
6	Cracking				Breaking long-chain alkanes into shorter hydrocarbons
7	Why cracking is carried out				Shorter chains are more in demand
8	Conditions for thermal cracking				High temperature (up to 1000 °C), high pressure (70 atm)
9	Main products of thermal cracking				Alkenes
10	Conditions for catalytic cracking				Zeolite catalyst, about 450 °C, slight pressure

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Combustion and Pollution **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Main products of catalytic cracking				Branched alkanes, cycloalkanes, aromatics
2	Products of complete combustion				$CO_2 + H_2O$
3	Products of incomplete combustion				CO and/or C, + H_2O
4	Why CO is toxic				Binds to haemoglobin
5	Pollutant formed from N_2 in engines				NO_x
6	Pollutant formed from sulfur impurities				SO_2
7	Problem caused by SO_2				Acid rain
8	Substance used to remove SO_2 from flue gases				CaO or $CaCO_3$
9	Metals in a catalytic converter				Pt, Pd, Rh
10	Catalytic converter equation for CO and NO				$2CO + 2NO \rightarrow 2CO_2 + N_2$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C12 Alkanes – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 3: Chlorination of Alkanes **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Why alkanes are unreactive				Strong, non-polar C–C and C–H bonds
2	Three greenhouse gases				CO_2, CH_4, H_2O
3	Condition for chlorination of methane				UV light
4	Type of reaction of methane with chlorine				Free-radical substitution
5	Radical				Species with an unpaired electron
6	Three stages of a free-radical reaction				Initiation, propagation, termination
7	Initiation equation for chlorine				$Cl_2 \rightarrow 2Cl\bullet$
8	Products of methane with chlorine (one substitution)				$CH_3Cl + HCl$
9	Why a mixture of products forms				Further substitution
10	Stage where two radicals combine				Termination

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.